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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE TITLE & CODE : DATA SCIENCE (24AD3107) A.Y. 2026 - 2027

Title of the project	Hand-Written Text Recognition using Deep Learning	
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Abstract

Handwritten text recognition is a valuable application of deep learning and computer vision, enabling computers to recognize and digitize handwritten text. This project describes a hybrid handwritten-note digitization system that utilizes an EMNIST-trained Convolutional Neural Network (CNN) alongside Microsoft TrOCR to identify handwritten text. EMNIST provides image classification at the character level using labelled 28 × 28 handwritten letters and digits while TrOCR is used to recognize complete handwritten text lines and enhance word-level readability. The EMNIST model learns visual patterns such as strokes, edges, shapes, and character structures while TrOCR employs an image Transformer encoder and text Transformer decoder. The system comprises image preprocessing, character and line segmentation, EMNIST-based classification, TrOCR-based line recognition, text reconstruction, confidence review, and PDF generation. This hybrid approach addresses the limitation of isolated-character classification when processing natural handwritten notes containing complete words and sentences. The recognized text can be displayed in an editable interface and exported as a downloadable PDF.

HoD-CSE Project Guide